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1.Introduction

The 2019 National Development Plan includes the Program for the Development of the Istmo de Tehuantepec, which promotes the regional Interoceanic Corridor known in Spanish as (CIIT). This project aims to improve connectivity, reduce transportation costs and times, and integrate the region into global logistics networks (SHCP, 2019; Torres Fragoso, 2022). However, the region has historically been a site of social conflict in response to incoming investments and development projects (Díaz Carnero, 2017). Although the government has anticipated an important development impact, it is crucial to analyze the perception of communities (Gobierno de México, 2021). Local perceptions directly influence the success of public policies and their adaptation to specific contexts (Fernández Moreno, 2008; Tavares-Martínez, Fitch-Osuna, 2019).

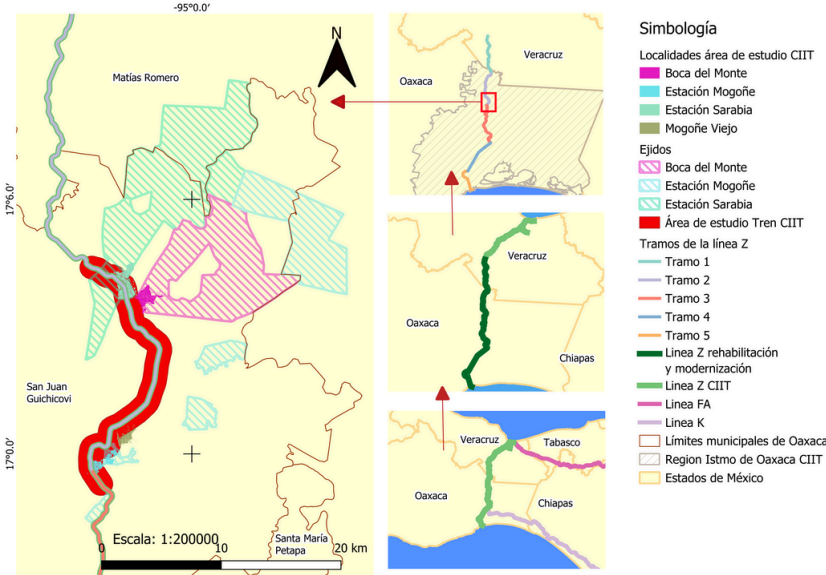


Figure 1. Location and sections of the CIIT indicating the study area. Created by the author based on GeoComunes, 2020, and INEGI and RAN, 2024.

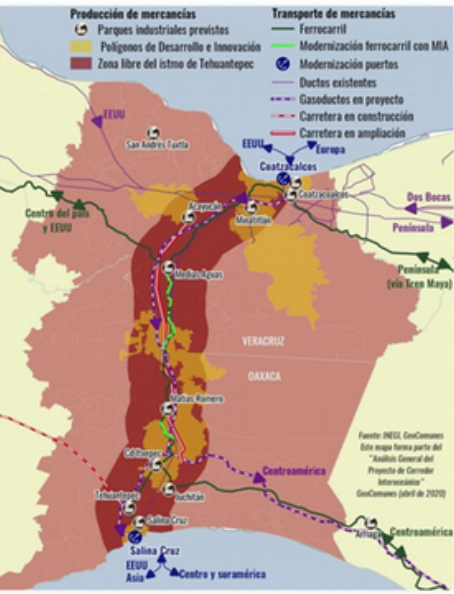


Figure 2. Map of the Interoceanic Corridor Megaproject of the Isthmus of Tehuantepec. Created by GeoComunes Collective, 2020.

2.Data and Methodology

This study uses a qualitative approach, applying Qualitative Comparative Analysis (QCA) and Social Network Analysis (SNA) to examine community perceptions of socio-environmental conflicts linked to the CIIT.

Fieldwork included 127 semi-structured and open interviews across four ejidos.

- Conflict characterization: Identification and mapping of socio-environmental conflicts in the Isthmus of Tehuantepec.
- Actor analysis: Identification, typology, and network mapping of social actors.
- Data collection: 120 semi-structured and 22 open interviews.
- Comparative analysis: Application of Qualitative Comparative Analysis (QCA).

3.Results

Mapping Socio-environmental Conflicts Related to the CIIT

A total of 15 socio-environmental conflicts related to the CIIT were identified in the Isthmus of Tehuantepec. The most frequent are intra-community disputes, followed by conflicts over industrial and energy megaprojects, and legal or territorial disputes linked to consultation and negotiation processes.

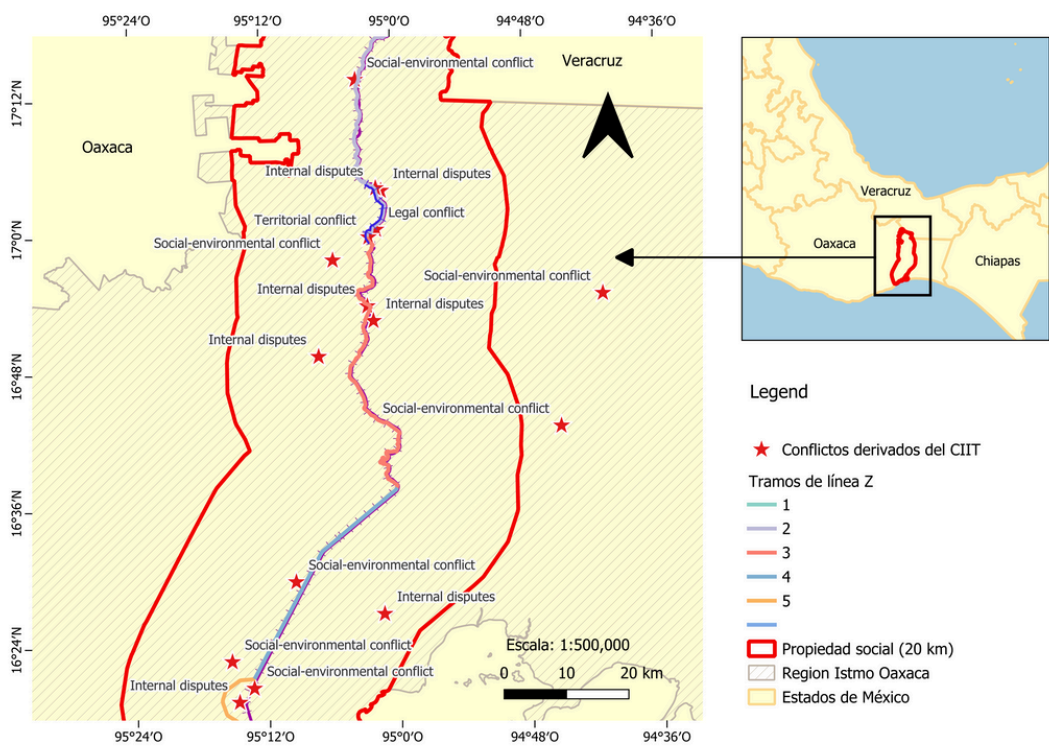


Figure 2. Map of Characterization and Identification of conflicts related to the CIIT. Created by the author based on GeoComunes, 2020, and INEGI and RAN, 2024.

5. Digital Humanities Innovative Research

Use of Technology and Artificial Intelligence

Different technologies and AI tools support the analysis of social perceptions related to the CIIT:

- Sentiment Analysis: NLP tools assess emotions and opinions from interviews and social media.
- AI-based GIS: Integrates spatial data to map conflicts and visualize land tenure dynamics.
- Predictive Modeling: Machine learning can helps forecast trends but not the whole socio-environmental perception.

References

Díaz Carnero, E. (2017). Energía eólica y conflictos socioterritoriales. El caso del Istmo de Tehuantepec, Oaxaca, México. Instituto de Geografía para la paz A.C.; Universidad Iberoamericana Santa Fe. *La electrificación y el territorio: historia y futuro*, 41, pp 1-21.

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<https://datos.gob.mx/busca/dataset/catalogo-de-nucleos-agrararios-de-la-propiedad-social>

3.1 Network of actors related to the CIIT

The network graph reveals two main centers of interaction: the CIIT, which concentrates most institutional and governmental connections, and UCIZONI, a non profit, which links social organizations and local communities. While the CIIT embodies the core of formal power and decision-making, UCIZONI represents a space for community organization and resistance to the megaproject. The different colors indicate clusters by type of actor, reflecting networks of cooperation, influence, and conflict. Overall, the network illustrates an unequal form of territorial governance where communities strive to balance top-down decisions imposed by institutional actors.

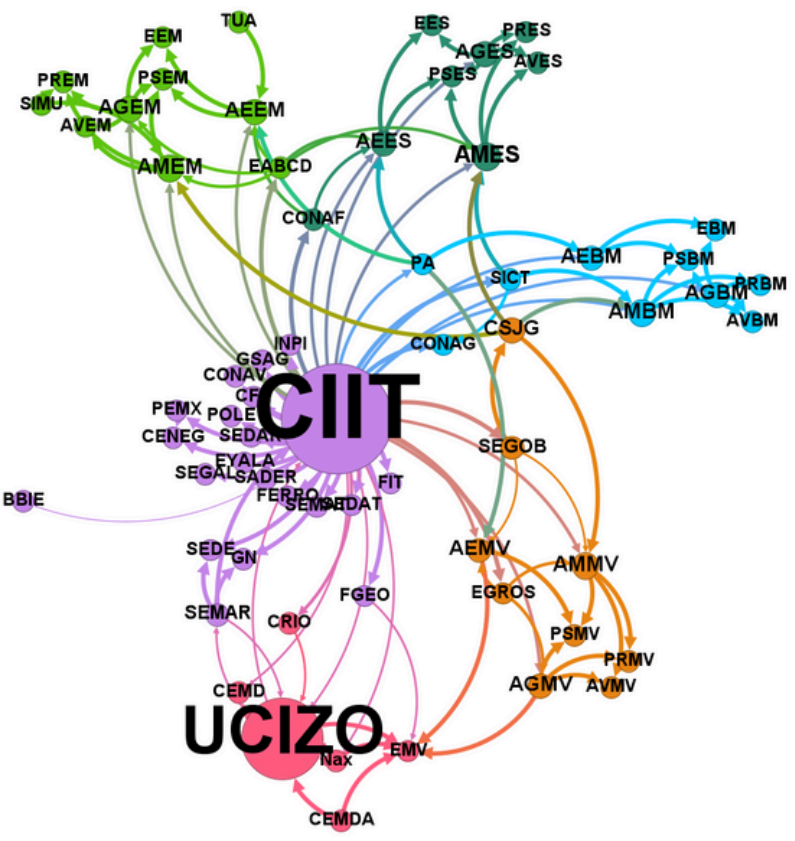


Figure 4. Network of actors involved in the implementation of the CIIT in four ejidos of the Isthmus of Oaxaca. Self elaboration based on the Local Actors Mapping (Tapella, 2023).

3.2 QCA logical map of the positive perception of the CIIT

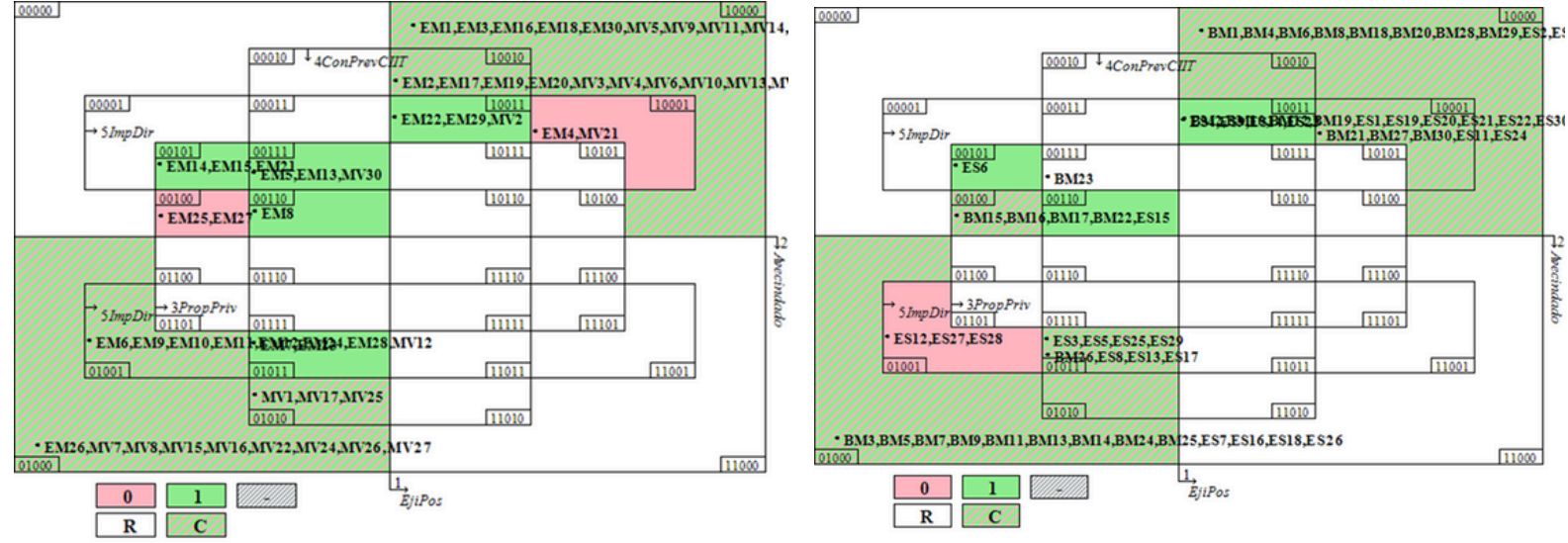


Figure 5. QCA logical map of the positive perception of the CIIT and the influence of land tenure in the ejidos of Estación Mogoñe and Mogoñe Viejo. Prepared using Tosmana 1.5 (Cronqvist, 2016).

Figure 6. Crisp-set QCA truth table on the influence of land tenure in the ejidos of Boca del Monte and Estación Sarabia. Prepared using Tosmana 1.5 (Cronqvist, 2016).

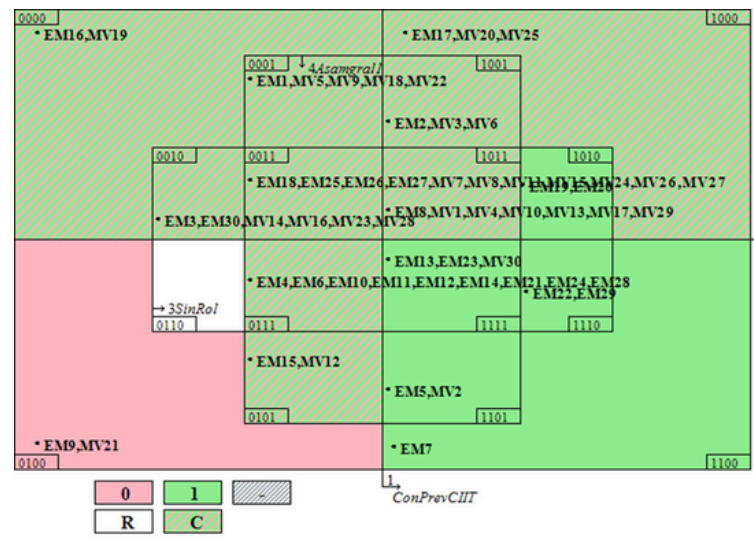


Figure 7. QCA logical map of the positive perception and the influence of territorial governance in the ejidos of Estación Mogoñe and Mogoñe Viejo. Prepared using Tosmana 1.5 (Cronqvist, 2016).

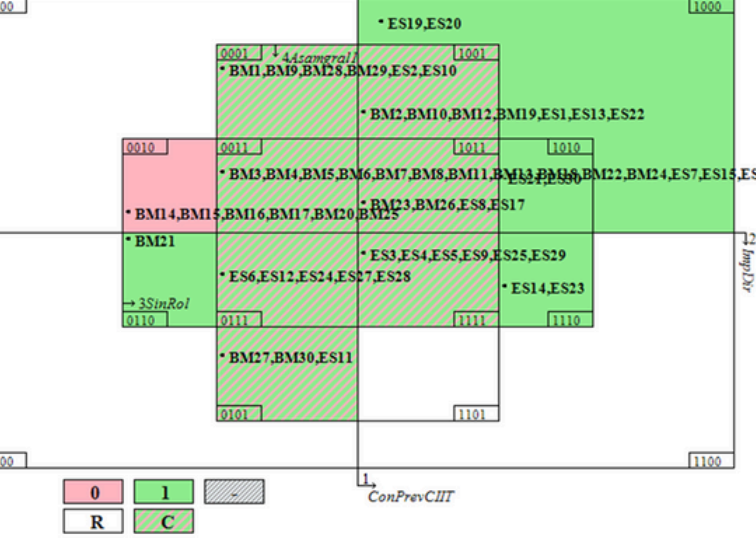


Figure 8. Crisp-set QCA logical map of the influence of governance in the ejidos of Boca del Monte and Estación Sarabia. Prepared using Tosmana 1.5 (Cronqvist, 2016).

QCA Logical Maps.

The analysis reveals that knowledge, public consultation, and direct benefits are key conditions for a positive perception of the CIIT. Land ownership acts as an influential variable, while territorial governance does not show a determining effect in the analyzed ejidos.

4. Conclusion

- In the collected narratives, development is primarily associated with infrastructure projects, such as railway modernization and road rehabilitation.
- According to the network of actors graphic, the Secretariat of the Navy (SEMAR) emerges as the most influential actor, holding the greatest power and strongest connections with the Interoceanic Corridor of the Isthmus of Tehuantepec (CIIT), as it directly oversees the coordination of the megaproject's development.
- The perception among the analyzed ejidos is generally positive toward the CIIT.
- Territorial governance does not appear to significantly influence this perception, since local communities have not participated in the planning or implementation stages of the project.
- Land tenure plays a role in shaping positive perceptions, particularly among landowners who have benefited from the construction works.
- Factors such as awareness of the project, prior consultation processes, and direct benefits on the communities are decisive in forming favorable perceptions toward the CIIT.

